

Eliminating the Auditor: Functional Inner Product Based Byzantine-Resilient Federated Learning

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Abstract—Privacy-Preserving Federated Learning (PPFL) is inherently vulnerable to Byzantine model poisoning, particularly when encrypted gradients prevent direct inspection. Existing defenses are often based on auditor-based architectures, introducing additional computational and communication overhead, as well as trust assumptions. This paper proposes an auditor-free Byzantine-resilient PPFL framework grounded in Functional Inner Product (FIP) geometry. By evaluating gradient alignment with a trusted reference direction and applying robust statistical thresholding, malicious updates are detected without decrypting raw gradients. Experimental results on MNIST under non-IID settings demonstrate superior robustness against targeted and untargeted poisoning attacks, achieving state-of-the-art targeted accuracy without external auditing.

Index Terms—Federated Learning, Privacy-Preserving, Functional Inner Product, Model Poisoning.

I. INTRODUCTION

Federated Learning (FL) is a scalable distributed optimization framework that enables collaborative model training without exposing raw data. By exchanging model updates instead of local datasets, FL enhances data locality and mitigates privacy risks. However, FL remains inherently vulnerable to model poisoning attacks, where Byzantine or adversarial clients manipulate gradients to disrupt convergence or induce targeted misclassification. This threat is further amplified in Privacy-Preserving Federated Learning (PPFL), where encrypted gradients prevent direct inspection and limit conventional anomaly detection.

Existing defenses rely on robust aggregation rules, statistical filtering, or auditor-based [1] architectures that evaluate gradient consistency prior to aggregation. Although auditor-centric frameworks improve robustness through clustering and distributional modeling, they introduce considerable computational and communication overhead. Furthermore, incorporating a trusted auditing entity increases cryptographic complexity, latency, and scalability as well as single-point-of-failure concerns in large-scale heterogeneous deployments.

To address these challenges, this research introduces an auditor-free Byzantine-resilient federated learning framework grounded in functional inner product analysis. By exploiting the geometric structure of client gradients in a functional inner product space to characterize alignment and coherence, adversarial updates are detected without external auditing modules. The resulting framework preserves privacy guarantees while reducing system overhead and maintaining robust convergence under adversarial participation.

Contributions: **1:** Auditor-free Byzantine-resilient FL with lightweight encrypted gradient alignment using functional

inner product geometry. **2:** Framework validated to improve robustness and scalability over auditor-based methods.

II. METHODOLOGY

The framework ensures Byzantine resilience without relying on an external auditor by embedding geometric consistency verification directly into the server-side aggregation process.

Assume a PPFL system with n clients and a central server, each client i computes a local gradient $g_i^{(t)} \in R^d$ at communication round t using model parameters $w^{(t)}$. To preserve privacy, gradients are encrypted using an additive homomorphic encryption scheme and transmitted as $E(g_i^{(t)})$. Instead of accessing raw gradients, the server evaluates their functional alignment with a trusted reference direction using the Functional Inner Product (FIP) [2].

Functional Inner Product (FIP): Given gradient functions g_i and g_j , their functional inner product over domain Ω is

$$\langle g_i, g_j \rangle = \int_{\Omega} g_i(x) g_j(x) dx, \quad (1)$$

which in discrete d -dimensional form becomes

$$\langle g_i, g_j \rangle = \sum_{k=1}^d g_i^{(k)} g_j^{(k)}. \quad (2)$$

Consistency of client i is evaluated via

$$\langle g_i^{(t)}, g_{\text{ref}}^{(t)} \rangle = \sum_{k=1}^d g_i^{(t,k)} g_{\text{ref}}^{(t,k)}. \quad (3)$$

Given gradients g_i and g_j , their functional inner product measures directional similarity and reduces to the standard dot product in discrete form. At round t , client consistency is evaluated via the projection $\langle g_i^{(t)}, g_{\text{ref}}^{(t)} \rangle$, where $g_{\text{ref}}^{(t)}$ denotes the trusted reference gradient representing the validated global descent direction. Significant deviation in this projection indicates directional misalignment and potential Byzantine behavior.

Reference Gradient $g_{\text{ref}}^{(t)}$: To establish an adversary-independent geometric anchor, the server computes an initial trusted gradient $g_{\text{ref}}^{(0)} = \nabla L(w^{(0)}; D_{\text{init}})$. Furthermore, For $t > 0$, the reference is updated recursively using momentum to ensure smooth adaptation while remaining robust to attacks: $g_{\text{ref}}^{(t)} = \beta \cdot g_{\text{ref}}^{(t-1)} + (1 - \beta) \cdot g_{\text{global}}^{(t-1)}$, where $\beta \in [0, 1]$ is the momentum factor that controls the trade-off between historical reference and new gradient information. This controlled update prevents abrupt directional drift caused by adversarial perturbations while preserving convergence stability.

Robust Statistical Bound for Malicious Client Detection.

To distinguish benign from malicious clients, alignment scores are evaluated using robust statistics. Let X denote the projection scores with median μ and dispersion estimated via Median Absolute Deviation (MAD). An adaptive acceptance interval $[\tau_{\min}, \tau_{\max}]$ is defined as $\tau_{\max} = \mu + 5 \text{MAD}$, $\tau_{\min} = \mu \cdot \alpha$ where $\alpha \in [0, 1]$ is selected based on the estimated attack rate $\rho^{(t)}$. The attack rate $\rho^{(t)}$ is computed as the proportion of clients whose alignment scores deviate significantly from the median relative to MAD.

A gradient is accepted if

$$\tau_{\min} \leq \langle g_i^{(t)}, g_{\text{ref}}^{(t)} \rangle \leq \tau_{\max}. \quad (4)$$

Gradients outside this interval are classified as malicious.

Secure Aggregation. Validated encrypted gradients are aggregated homomorphically: $E(g_{\text{global}}^{(t)}) = \sum_{i \in \mathcal{S}^{(t)}} E(g_i^{(t)})$, where $\mathcal{S}^{(t)}$ is the accepted client set. The decrypted result updates the global model.

III. EXPERIMENT & RESULTS

A. Experimental Setup

This study strictly follows the experimental setup of the base paper [1] for fair benchmarking, using logistic regression on MNIST under non-IID settings. The proposed framework operates on homomorphically encrypted gradients and is evaluated against both targeted and untargeted poisoning attacks. Furthermore, performance is measured using Target Accuracy (T_{acc}), Other Label Accuracy (O_{acc}), and Overall Accuracy (ACC) to quantify attack success and defense robustness.

B. Performance Evaluation

Table I presents the results under the Non-IID setting (with attack rate = 50%). Without defense, targeted attacks severely degrade performance, reducing T_{acc} to 0.07 despite moderate overall accuracy. With the proposed defense, T_{acc} improves significantly to 0.78 while maintaining high Other Label Accuracy (0.97) and strong overall performance (0.97). These results demonstrate effective mitigation of targeted poisoning in Non-IID settings.

TABLE I
EVOLUTION OF ACCURACY UNDER IID AND NON-IID SETTINGS

Without Defence			Proposed Defence		
Targeted		Untargeted	Targeted		Untargeted
T_acc	Oracy	ACC	T_acc	Oracy	ACC
0.07	0.89	0.81	0.97	0.97	0.88

Figure 1 illustrates the impact of increasing malicious client rates on model performance, showing target class accuracy under targeted attacks (right) and overall accuracy under untargeted attacks (left).

C. Comparison with Existing Methods

Table II compares the proposed model with existing schemes, whose results are referenced from the base paper [1]. The proposed method achieves the highest targeted accuracy (97.69%) and competitive untargeted accuracy (88.82%), demonstrating strong robustness without an external auditor.

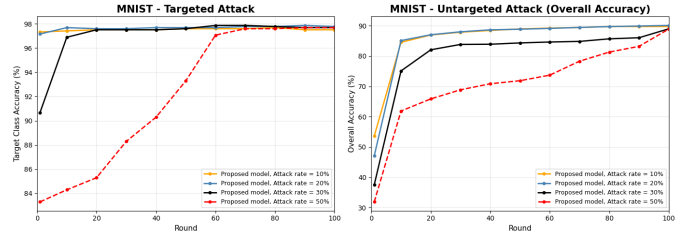


Fig. 1. Performance with targeted and untargeted attack and non-IID data.

TABLE II
ACCURACY COMPARISON BETWEEN EXISTING SCHEMES

Work	Attack	
	Targeted	Untargeted
Krum	71.4	78.3
Auror	32.5	49.9
PEFL	46.4	57.4
Sybils	89.7	89.5
FL-trust	37.8	43.4
ShieldFL	90.1	89.4
Auditor-Based [1]	96.5	96.3
Proposed model	97.69	88.82

As shown in Table III, compared to the baseline auditor-based method, which incurs quadratic complexity due to clustering and covariance computations, the proposed auditor-free approach uses linear-complexity inner product analysis, removes extra communication hops, and achieves scalable, efficient performance with respect to both client count and model dimension.

TABLE III
COMPLEXITY COMPARISON

Component	Auditor-Based [1]	Proposed
Clustering (GMM)	$\mathcal{O}(nd^2)$	—
Distance Computation	$\mathcal{O}(nd^2)$	—
Inner Product Analysis	—	$\mathcal{O}(nd)$
Extra Communication Hop	Yes	No
Scalability	Limited	Linear (n, d)

IV. CONCLUSION

This work presents an auditor-free, Byzantine-resilient federated learning framework that leverages functional inner product geometry and robust statistical thresholding to detect malicious updates in encrypted gradients. Experiments on MNIST demonstrate strong robustness against both targeted and untargeted poisoning attacks, maintaining high accuracy while reducing computational and communication overhead. Future work will evaluate the framework on IID settings and a wider variety of datasets to further validate its scalability and effectiveness.

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